

99 — Source Coverage Ledger

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Revision: 1 Last modified: 2026-07-05T14:20:00Z **Status:** Draft
— consolidated from 99-source-coverage-ledger.md.

1. Position

This section is the §11.4.118 proof-of-completeness for the MVP documentation consolidation. It maps every canonical source document in docs/research/mvp/final/ to the implementation section that carries its distinctive content, and it surfaces any source ideas that are not yet fully reflected.

The authoritative version lives at [../99-source-coverage-ledger.md](#); this README is the navigable consolidation.

2. Source → implementation mapping

Source	Implementation section(s)	Distinctive contribution
SPECIFICATION.md	<u>00 — Executive Summary</u> , <u>02 — System Architecture</u>	Architectural spine, roles, principles, roadmap, decision register
00-product-scope-and-principles.md	<u>01 — Product Scope</u>	Product definition, personas, scope, parity matrix, decisions D1–D8
01-data-plane.md	<u>03 — Data Plane</u>	Transport trait, transports, routing, multihop, MTU, DAITA

Source	Implementation section(s)	Distinctive contribution
02-control-plane.md	<u>04 — Control Plane,</u> <u>08 — API Contracts</u>	Go monolith, data model, events, coordinator, API, reconciliation
03-client-core-and-ui.md	<u>05 — Client Core & UI</u>	FFI surface, Flutter UI, platform shims, state management
04-security-privacy-pki.md	<u>06 — Security, Privacy & PKI</u>	Zero-trust, identity, PKI, no-logging, kill-switch, PQ, threat model
05-repo-layout-tooling-and-helix-ecosystem.md	<u>07 — Infrastructure & DevOps</u>	Repo layout, codegen, helixvpnctl, substrates, ecosystem
06-phase0-spike-wbs.md	<u>11 — Guides & FAQs</u>	Phase-0 gates G1-G6, milestones S0-S8
07-phase1-mvp-wbs.md	<u>11 — Guides & FAQs</u>	Phase-1 MVP WBS, 8-criteria DoD, SLOs
08-phase2-parity-wbs.md	<u>11 — Guides & FAQs</u>	Phase-2 parity WBS, P2-AC + SLOs
09-phase3-reach-wbs.md	<u>11 — Guides & FAQs</u>	Phase-3 extended-reach WBS, gates G20-G26
10-testing-acceptance-and-qa.md	<u>09 — Testing & QA</u>	§11.4.169 test types, evidence model, acceptance gates
11-deep-research-appendix.md	<u>12 — Appendix: Research</u>	Consolidated cited research (10 angles)
99-source-coverage-ledger.md	this section	Source-coverage proof and residual gaps
MASTER_INDEX.md	<u>00 — Executive Summary</u>	Full document tree, statuses, quality items
REFINEMENT_NOTES.md	<u>00 — Executive Summary</u>	Pass-1→pass-N punch-list

Nano-detail volumes

Volume	Dir	Implementation section
V0 Meta	v00-meta/	<u>00</u> , this section
V1 Product	v01-product/	<u>01</u> , <u>08</u>
V2 Data Plane	v02-data-plane/	<u>03</u>
V3 Control Plane	v03-control-plane/	<u>04</u> , <u>08</u>
V4 Clients	v04-client/	<u>05</u>
V5 Security	v05-security/	<u>06</u>
V6 Deploy	v06-deploy/	<u>07</u>
V7 Execution	v07-execution/	<u>11</u>
V8 Testing	v08-testing/	<u>09</u>
V9 Research	v09-research/	<u>12</u>
V10 Design	v10-design/	<u>10</u>

External design-system index

File	Implementation section
docs/design/README.md	<u>10 — Design System</u>

3. Residual coverage gaps

The following distinctive source ideas are deliberately not adopted or are recorded as open.

G1 — [03_ZAI] disaster-recovery completeness — RESOLVED

v06-deploy/disaster-recovery.md now consolidates the RTO/RPO budget, KMS-encrypted backups, and Terraform-driven region-failover runbook that the original ledger called for. Independently re-verified 2026-07-04. No further action.

Traceability note: This resolves the documentation side of requirements-traceability **GAP-3** (NFR-205 DR RTO/RPO targets are still UNVERIFIED pending the first **CHAOS region-failover drill in Phase 2** — a measurement deferral, not a missing doc).

G2 — [06_GRK] sing-box transport framework — deliberate divergence

Proposed: use sing-box as the universal transport multiplexer.

Decision: not adopted. The protocols it carries (Hysteria2, Shadowsocks, UDP-over-TCP) are reflected in the Phase-2 transport set, but the framework itself conflicts with the Rust-core decision and the iOS NE memory ceiling. The custom Rust helix-transport crate gives byte-for-byte client↔edge reuse.

Rationale recorded: 99-source-coverage-ledger.md §G2 and 03 — Data Plane.

G3 — [09_GCT] Fyne desktop UI — deliberate divergence

Proposed: use Fyne for desktop/web client. **Decision: not adopted.** Flutter reaches all 8 required platforms (including Aurora OS and HarmonyOS NEXT) with one codebase and a shared helix_design system. Fyne lacks mobile/niche-OS reach.

Rationale recorded: 99-source-coverage-ledger.md §G3 and 05 — Client Core & UI.

GAP cross-check (requirements-traceability.md register)

The independent GAP-1..GAP-6 register in v00-meta/requirements-traceability.md is kept current with this consolidation:

GAP	Status	Residual
GAP-1	Closed in source docs	local-ACL × central-policy precedence pinned in v03-control-plane/svc-policy.md and v04-client/helix-core-rust.md
GAP-2	Intentional / documented	FR-1103 (Rosenpass) is an evaluation, not a runtime capability; no runtime test type required
GAP-3	Doc-level closed	NFR-205 RTO/RPO targets remain UNVERIFIED until the first Phase-2 CHAOS region-failover drill measures them
GAP-4	Closed in source docs	Single owning doc created: v04-client/connector.md consolidates FR-701..707 and the advertise/route/registry contracts
GAP-5	Closed at doc level	

GAP	Status	Residual
GAP-6	Closed in source docs	All 15 v08-testing/ docs + coverage-ledger-schema.md authored; evidence states are PENDING in spec phase NFR-413/NFR-414/FR-610 added to pin DDOS/RBAC/rate-limiting owners; quantitative targets remain UNVERIFIED until Phase-2 benchmarks

4. Decision coverage cross-check

All eight key decisions are surfaced explicitly in final/:

Decision	Camp A	Camp B	Surfaced in
D1 obfuscation	MASQUE/ QUIC	Hysteria2 + Salamander	01-data-plane.md, 08-phase2-parity-wbs.md, 11-deep-research-appendix.md
D2 client core	Rust	Go	03-client-core-and-ui.md, 11-deep-research-appendix.md
D3 event bus	Redis → NATS	NATS from start	02-control-plane.md, 08-phase2-parity-wbs.md
D4 subnet collision	ULA/4via6	CGNAT	02-control-plane.md, 08-phase2-parity-wbs.md
D5 edge language	Rust	Go	01-data-plane.md, 06-phase0-spike-wbs.md
D6 transport topology	single protocol	asymmetric per-leg	01-data-plane.md, 11-deep-research-appendix.md
D7 ambition	lean	Connectivity-OS	00-product-scope-and-principles.md, SPECIFICATION.md
D8 licensing	source-available + commercial	pure OSS	SPECIFICATION.md §9

5. Cross-references

- Authoritative ledger → [../../../../99-source-coverage-ledger.md](#)
- Synthesis evidence base → [../../../../v09-research/_SYNTHESIS.md](#)
- Gap analysis → [../../../../reviews/mvp-final/findings/phase1-docs-gap-analysis.md](#)

Source: docs/research/mvp/final/99-source-coverage-ledger.md.